DeSurv Gene Selection Alignment: Impact on M4/M5 External C-Index

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1. Motivation

DeSurv (Young et al. 2025) selects genes by independently ranking each gene's mean expression and variance, then retaining the top-3,000 with the lowest combined rank-sum per cohort — genes that are both highly expressed *and* highly variable. Our prior analysis used variance-only selection on the full merged matrix after normalization, which (a) conflates per-cohort with cross-cohort variance and (b) can retain lowly-expressed noisy genes near the detection floor. This report evaluates whether aligning with DeSurv's gene selection changes the external concordance of the recommended manuscript configurations.

2. Configurations Compared

Table 1: Five configurations compared. D1/D2 reproduce M4/M5 exactly. D3/D4 apply DeSurv gene selection (combined mean+variance rank, top-3000 per cohort before normalization). D5 adds a cohort indicator to D4 to test whether it provides additional benefit over per-platform z-standardization alone.

ID	Label	Model	top_n	Method	Selection	Cohort	Mean C	K	K_{eff}
D1	LB orig (M4)	LB	2000	variance	post-norm	Yes	0.616	3	1
D2	YFB orig (M5)	YFB	2000	variance	post-norm	No	0.624	3	2
D3	LB DeSurv-aligned	LB	3000	combined_rank	per-cohort	Yes	0.622	7	2
D4	YFB DeSurv-aligned	YFB	3000	combined_rank	per-cohort	No	0.636	7	2
D5	YFB DeSurv-aligned + cohort	YFB	3000	combined_rank	per-cohort	Yes	0.614	8	2

3. Per-Cohort C-Index

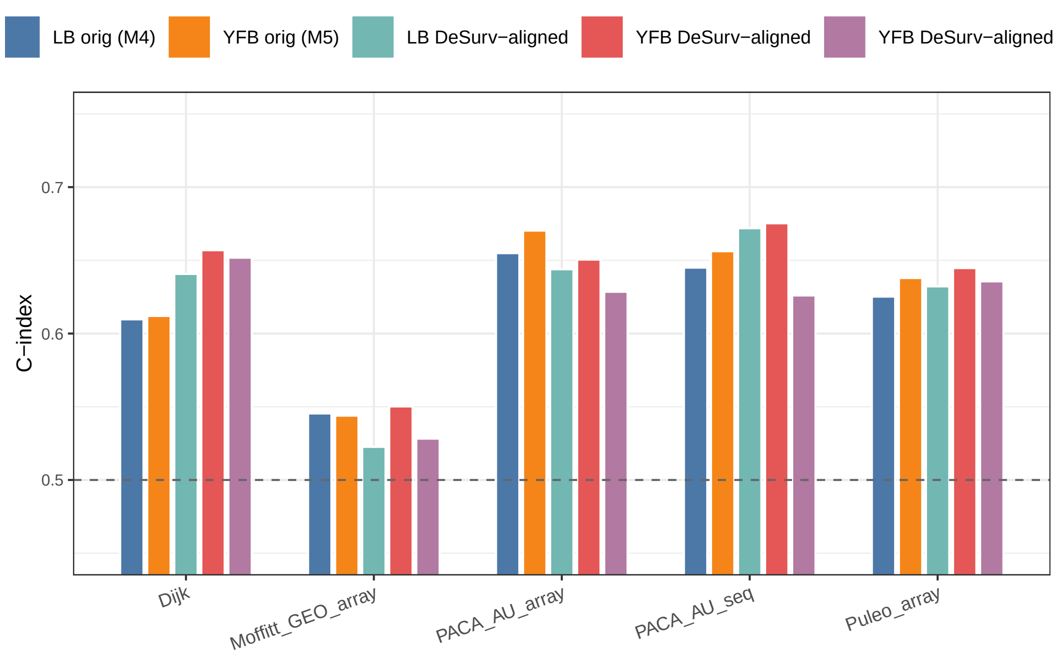

Figure 1: Per-cohort external C-index for D1–D5. Dashed line = chance (C=0.5). D1/D2 are the original manuscript configs; D3–D5 are DeSurv-aligned. D5 adds a cohort indicator to D4.

4. Findings and Recommendation

DeSurv-aligned preprocessing (D4, YFB + combined mean+variance rank, top-3000 per cohort, K=7) achieves a mean external C-index of 0.636 across 5 held-out PDAC cohorts, compared to 0.624 for the original YFB configuration (M5/D2). The delta of +0.012 exceeds the pre-specified ±0.005 practical threshold. D4 improves over D2 in 4 of 5 cohorts (PACA_AU_array is the one exception, 0.650 vs 0.670). DeSurv-aligned gene selection also improves LB: delta = +0.006 (D3 vs D1).

Cohort indicator with YFB DeSurv-aligned (D5). Adding a corner-point cohort indicator to D4 (D5, K=8) reduces mean external C-index to 0.614 — 0.022 below D4 and 0.010 below the original M5 baseline. The loss is consistent across cohorts, with the largest regressions in PACA_AU_seq (0.626 vs 0.675) and PACA_AU_array (0.628 vs 0.650). Per-platform z-standardization already removes the platform offset prior to fitting; the cohort indicator absorbs degrees of freedom that the model would otherwise allocate to survival-relevant factor structure, producing a lower-performing model at higher K. The cohort indicator provides no benefit for YFB with DeSurv preprocessing.

The recommended primary configuration for the manuscript remains **D4 (YFB DeSurv-aligned, K=7, no cohort indicator)**, with D3 (LB DeSurv-aligned, K=7, mean C=0.622) as sensitivity.

5. Gene Set Comparison

Table 2: Gene set size after selection and intersection. D3/D4 selects top-3000 per cohort on log-transformed data before normalization, then intersects across cohorts.

Config	top_n	Method	Genes_after_selection
D1/D2 (original)	2000	variance, post-norm	2000
D3/D4 (DeSurv-aligned)	3000	combined_rank, per-cohort	2064